

How should we be performing time series simulation studies?

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Time Series Connect



- Why do we conduct simulation studies?
- How do we (as a field) currently construct them?
- My hypothesis as to why it evolved like that.
- Can we do any better?

Recently published in The American Statistician ¹

¹<https://doi.org/10.1080/00031305.2026.2695128>

Why sim studies?

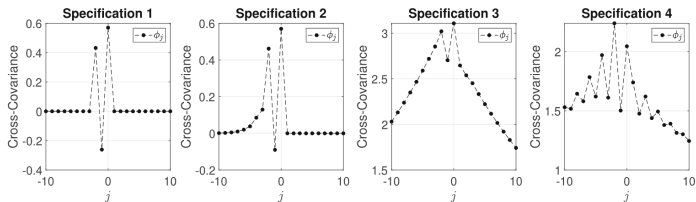
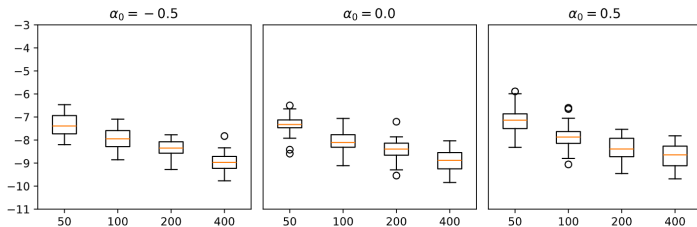
- Backup theoretical results with practical results?
- Provide results where we don't have theory?
- Evaluate finite sample properties?
- Stress-test assumption violations?
- Benchmark against competitors?
- Validate real-world findings in matched simulations?

Why sim studies?

- Backup theoretical results with practical results?
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- Because my advisor told me to
 - Because I need to in order to publish
 - Because reviewers asked for it

Not trying to nit pick but JTSA is the journal for the field

- JTSA published 44 papers in 2024, 39 had an AR simulation
- Only 2 papers selected coefficients randomly (in $AR(1)$)
- Of the remaining 37, 25 used a single AR parameter setting!
- Only 12/39 considered orders larger than 1!



Specific parameter values, 7,2,2

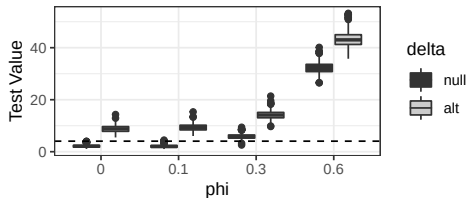
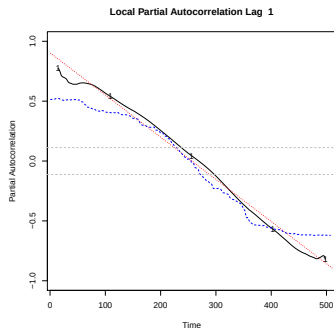
Model 7

$$X_t = \log(4 \cdot e^{0.9 \cdot X_{t-2}} + 5 \cdot e^{0.9 \cdot X_{t-1}} + 6 \cdot e^{0.9 \cdot X_{t-3}}) + \epsilon_t$$

estimation results for MthINGARCH(1,1) series with $n = 1000$, $m = 6$, $\theta_0 = (0.3, 0.4, 0.2)'$,

$$\epsilon_t \sim \mathcal{NB}(1, \frac{1}{2}), \text{ and } \sigma_0^2 = 2$$

Table IV. Bias and variance of parameter estimates in system (20)



- Simulation across the AR whole space is hard
 - AR(1) is easy but we like to cherry pick
 - AR(2) has 50% random chance, AR(3) is 1/12
- Beyond AR is often harder
- We let each other get away with it - citing precedent

What is the whole AR space?

- Want to simulate across the causal space
- For a wide set of AR, MA and ARMA processes

BUT HOW?

AR(p):

$$Y_t = \sum_{j=1}^p \phi_j Y_{t-j} + \epsilon_t \quad t = 0, \pm 1, \pm 2, \dots,$$

Causal:

$$Y_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j} \quad \sum_{j=0}^{\infty} |c_j| < \infty;$$

How to check causality?

AR(p):

$$Y_t = \sum_{j=1}^p \phi_j Y_{t-j} + \epsilon_t \quad t = 0, \pm 1, \pm 2, \dots,$$

Causal:

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How to check causality?

Roots of the polynomial:

$$\Phi(x) = 1 - \phi_{p,1}x - \phi_{p,2}x^2 - \dots - \phi_{p,p}x^p$$

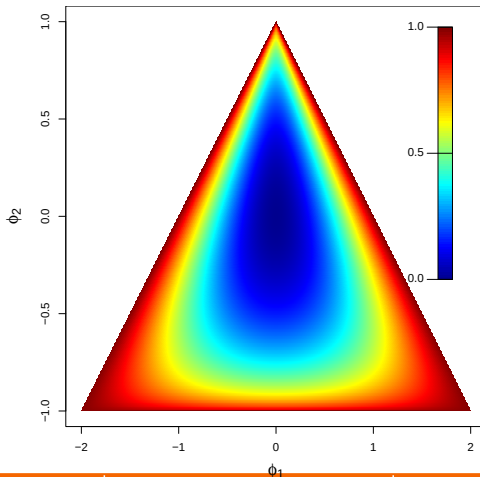
Are the roots outside the unit circle?

Known solution spaces

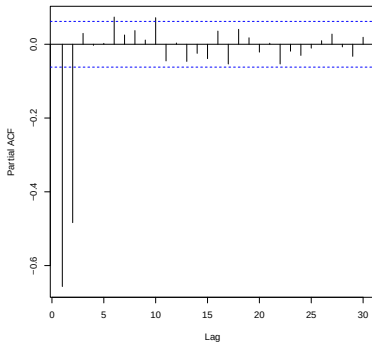
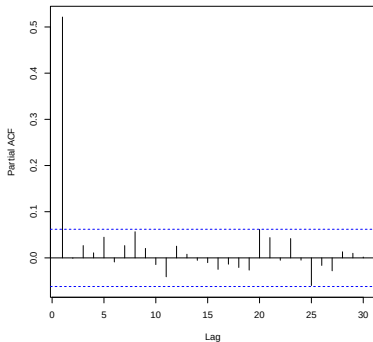
AR1: $\phi_{1,1} \in (-1, 1)$

AR2:

What about higher orders?



PACF reminder



PACF at lag h : $\alpha(h)$.

$$|\alpha(h)| < 1 \quad \text{for} \quad 1 \leq h \leq p \quad \text{and} \quad \alpha(h) = 0 \quad \text{for} \quad h > p.$$

Ramsey (1974) noted a 1:1 mapping. Any point in $(-1, 1)^p$ corresponds to a unique point in the causal parameter space:

$$(\phi_{p,1}, \dots, \phi_{p,p}) = g(\alpha(1), \dots, \alpha(p))$$

$$(\alpha(1), \dots, \alpha(p)) = g^{-1}(\phi_{p,1}, \dots, \phi_{p,p}).$$

Note: All of the causal boundary points are where one or more correlations are equal to 1 or -1.

The map g can be described via the Durbin-Levinson algorithm, which provides

$$\phi_{k,k} = \alpha(k)$$

$$\phi_{k,j} = \phi_{k-1,j} - \phi_{k,k}\phi_{k-1,k-j} \quad k > 1, 1 < j < k - 1.$$

Equivalent approaches

Let R_p denote the causal parameter space for an order p AR. Any of the following conditions can be used to assess if $\phi \in R_p$.

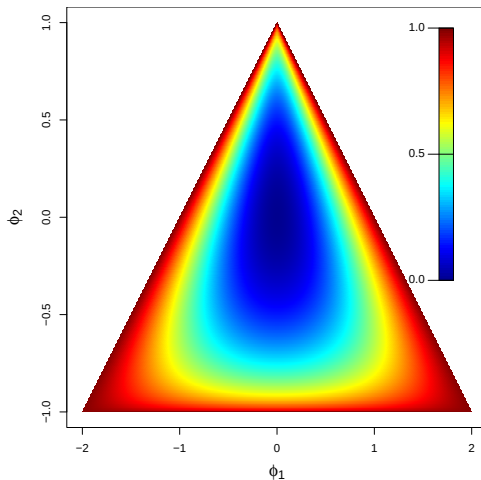
1. The roots of $\Phi(x) = 1 - \phi_{p,1}x - \phi_{p,2}x^2 - \cdots - \phi_{p,p}x^p$ are outside the unit circle.
2. $|\phi_{h,h}| < 1$ for $1 \leq h \leq p$, and for each $k = 2, \dots, p$ and $j = 1, \dots, k$, $\phi_{k,j} = \phi_{k-1,j} - \phi_{k,k}\phi_{k-1,k-j}$.
3. $|\phi_{p,p}| < 1$, $\phi_{2i,2i} > -1$ for $i < p/2$, and for each $k = 2, \dots, p$ and $j = 1, \dots, k$,

$$\sum_{j=1}^k \phi_{k,j} < 1 \quad \text{and} \quad \sum_{j=1}^k (-1)^{j+1} \phi_{k,j} > -1,$$

with $\phi_{k,j} = \phi_{k-1,j} - \phi_{k,k}\phi_{k-1,k-j}$.

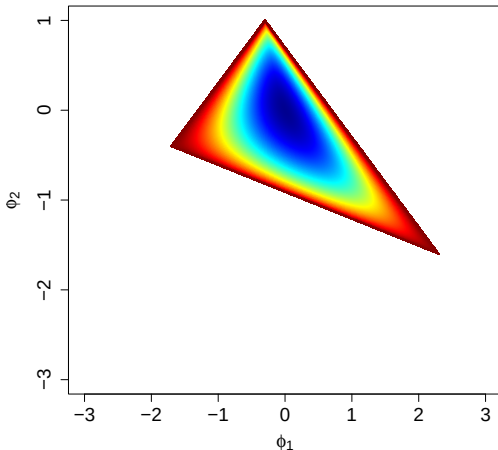
4. For $p > 1$, $(\phi_{p-1,1}, \dots, \phi_{p-1,p-1}) \in R_{p-1}$, $|\phi_{p,p}| < 1$ and $\phi_{p,j} = \phi_{p-1,j} - \phi_{p,p}\phi_{p-1,p-j}$ for $j = 1, \dots, p-1$.

$$|\phi_{2,2}| < 1 \quad \text{and} \quad |\phi_{2,1}| < 1 - \phi_{2,2}$$

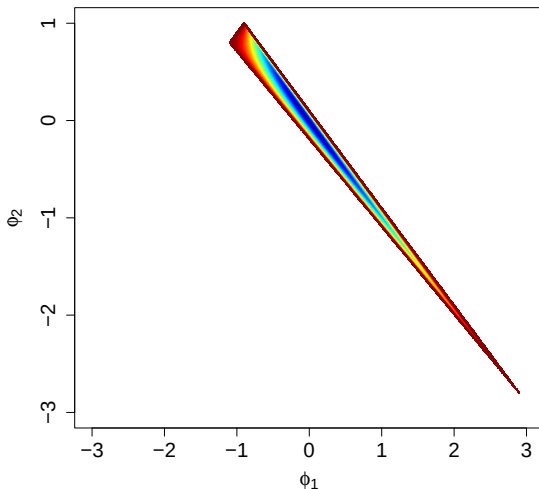


$$|\phi_{3,3}| < 1, \quad |\phi_{3,1} + \phi_{3,3}| < 1 - \phi_{3,2}, \quad \phi_{3,2} > -1 + \phi_{3,3}(\phi_{3,3} - \phi_{3,1})$$

Degree of Correlation: AR(3), $\phi_3 = 0.3$

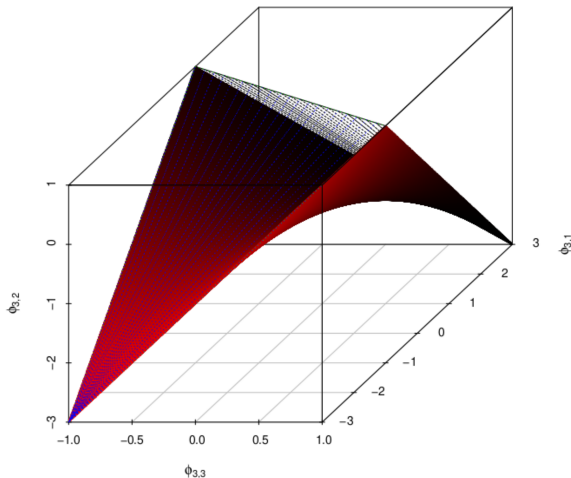


Degree of Correlation: AR(3), $\phi_3 = 0.9$

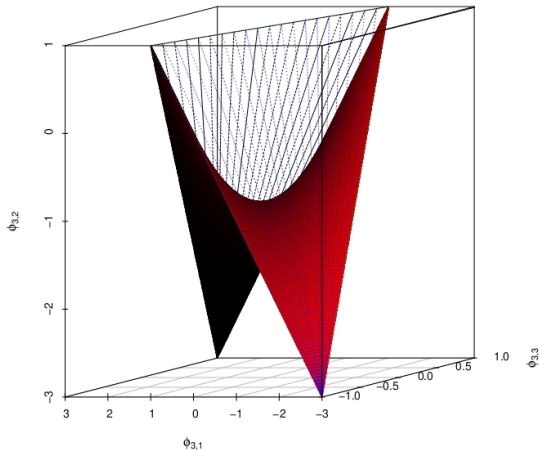


Causal Region AR3

$$|\phi_{3,3}| < 1, \quad |\phi_{3,1} + \phi_{3,3}| < 1 - \phi_{3,2}, \quad \phi_{3,2} > -1 + \phi_{3,3}(\phi_{3,3} - \phi_{3,1})$$



Causal Region AR3



Dropbox Link

So what?

Need to summarise: $R_n(i, j) = \text{corr}(Y_i, Y_j)$, for $1 \leq i, j \leq n$

Measure of total correlation:

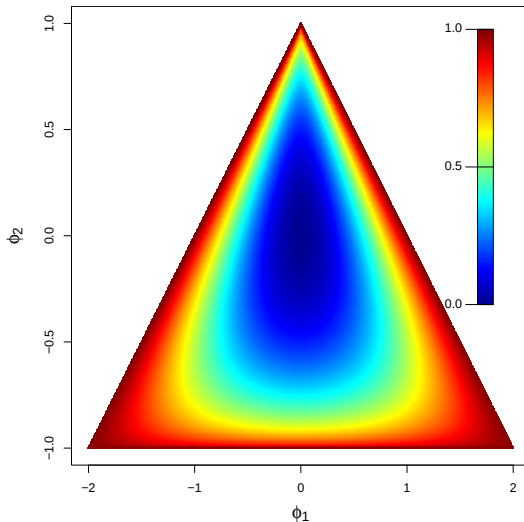
$$\rho(\{Y_t\}) = 1 - \det(R_n) = 1 - \prod_{k=1}^p (1 - \alpha(k)^2)^{p+1-k}.$$

where $\det$ denotes the determinant.

$\rho(\{Y_t\}) \rightarrow 1$, high correlation

$\rho(\{Y_t\}) \rightarrow 0$, low correlation

AR2 Total Correlation

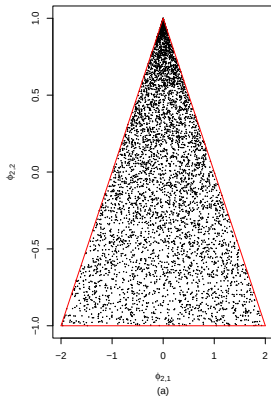
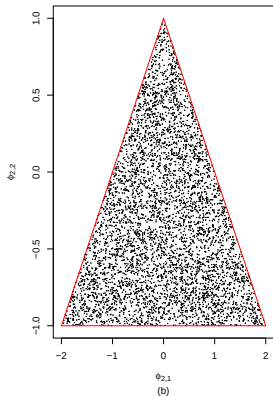
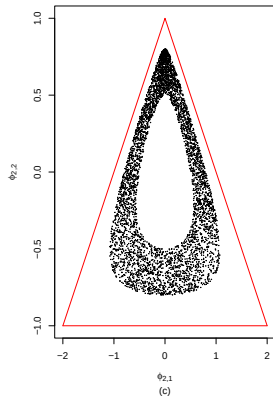


1. Generate $\{X_i\}_{i \in (1, \dots, p)}$ independently having, $0 \leq \ell, u \leq 1$
Uniform density

$$f(x) = \frac{1}{(u - \ell)} \mathbb{I}_{\ell < x < u}.$$

2. Generate $\{R_i\}$ independent of $\{X_i\}$,
 $P(R = 1) = P(R = -1) = 0.5$.
3. Let each $\alpha(i) = R_i X_i$ and solve for $\{\phi_{p,j}\}$.

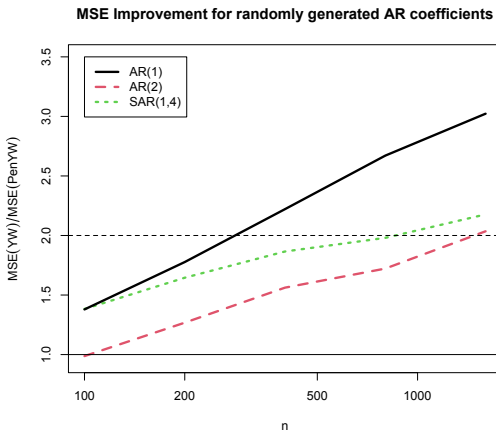
Larger values of (ℓ, u) for one or more i correspond to more correlation.

Uniform in the α SpaceUniform in the ϕ SpaceUniform in p ; $0.5 < p < 0.8$ 

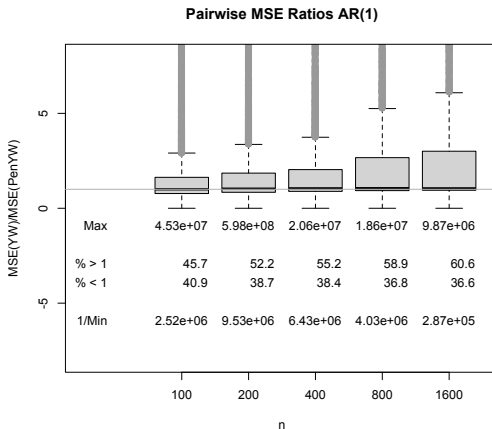
How do we use this?

MSE for YW AR(p) fits based on $\hat{\pi}$ vs $\tilde{\pi}$, 10,000 replications.

Parameters randomly generated over parameter space.

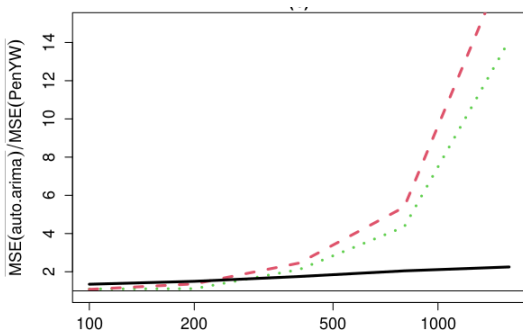


Pairwise MSE for YW AR(1) fits based on $\hat{\pi}$ vs $\tilde{\pi}$.



MSE for YW AR(p) fits based on $\hat{\pi}$ vs $\tilde{\pi}$, 10,000 replications

Parameters randomly generated over parameter space.



Provable consistency!

Model	Method	$n = 100$	$n = 200$	$n = 400$	$n = 800$	$n = 1600$
AR(1)	PenYW	0.78	0.84	0.88	0.91	0.94
	YW	0.68	0.69	0.70	0.69	0.69
	auto.arima	0.69	0.71	0.72	0.72	0.72
AR(2)	PenYW	0.78	0.84	0.89	0.90	0.93
	YW	0.68	0.69	0.71	0.72	0.69
	auto.arima	0.72	0.75	0.77	0.76	0.73
SAR(1,4)	PenYW	0.71	0.78	0.84	0.90	0.93
	YW	0.63	0.65	0.66	0.68	0.70
	auto.arima	0.70	0.77	0.76	0.80	0.81

- Showed how PACF and AR coefficients are linked
- Used this link to describe causality regions in higher dimensions
- Visualized up to AR(4)
- Utilized the new description for simulation study generation

A call to arms - we should enforce a new standard of better simulation studies

Code available in supplementary material to the TAS paper:

<https://doi.org/10.1080/00031305.2026.2695128>